

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456785.6871 +/- 0.0007 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.015
Transit depth (Rp/Rs)^2	3.04 +/- 0.51 [%]
Orbital Inclination (inc)	81.93 +/- 1.25
Airmass coefficient 1 (a1)	1.379 +/- 0.017
Airmass coefficient 2 (a2)	-0.1781 +/- 0.0069
Scatter in the residuals of the lightcurve fit is	1.26 %
Best Comparison Star	None
Optimal Aperture	2.83
Optimal Annulus	3.91
Transit Duration (day)	0.0512 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.015 vs 0.1594 (deviation 0.97σ)
Duration fit vs published	0.0512 d vs 0.0512 d (deviation 0.01σ)
Mid-time O–C	1.3 min
Depth SNR	11.6
Residual scatter	1.26 %

Light curve

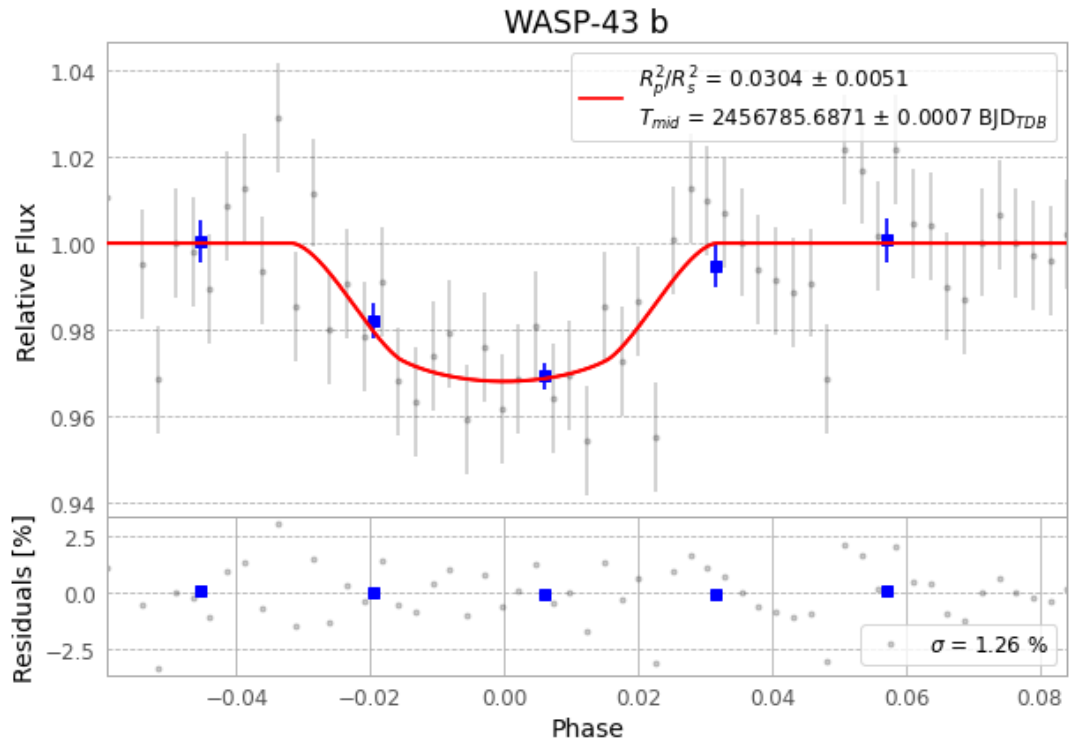


Figure 1 — FinalLightCurve_WASP-43 b_2014-05-07.png (SHA-256 29f17499c0d24939...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [320, 244], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1876eda07e0d27f8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

57 FITS frames (37 MB), WASP-43140508031512.FITS → WASP-43140508060312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140508.log (ddc5b887776d...), FinalLightCurve_WASP-43 b_2014-05-07.png (29f17499c0d2...), FinalLightCurve_WASP-43 b_2014-05-07.csv (09c9ea48a27e...)

Compute resources

Wall time 112.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 23:30Z.